THE OBSTRUCTION CLASS FOR DEFORMING A MORPHISM OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the situation and the equation-to-solve fixed in the preceding section of the chapter; the proof recalls the two identifications it uses.

Editorial note. The eight references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The identification of the naive cotangent complex with the truncation of the cotangent complex for ringed topoi. Lemma lemma-compare-cotangent-complex-with-naive-ringed-topoi, used at proof lines 4046-4048. Transfers the deformation-theoretic statements.
- The identification H^0(L_f) = Omega_f for a morphism of ringed topoi. Lemma lemma-H0-L-morphism-ringed-topoi, used at proof lines 4048-4050. Is the second identification needed for parts (2) and (3).
- The deformation-theory obstruction, parametrization and automorphism statements for ringed topoi. Deformation Theory, Lemma defos-lemma-parametrize-solutions-ringed-topoi, Lemmas defos-lemma-huge-diagram-ringed-topoi and defos-lemma-choices-ringed-topoi, used at proof lines 4050-4052 and 4059-4061. Supply parts (2), (3) and the class xi'.
- The distinguished triangle for the pair of morphisms of ringed topoi. Lemma lemma-triangle-ringed-topoi, used at proof lines 4080-4091. Gives the long exact sequence whose image is the obstruction.

Source reference. Original lemma 08UK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11203–11203. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08UG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11199–11199. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0GQ9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20854–20854. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 08UE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11197–11197. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 08V4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11222–11222. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08V0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11218–11218. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08V2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11220–11220. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08V1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11219–11219. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *In the situation above we have*

(1) *There is a canonical element* $\xi \in \mathrm{Ext}^2_{\mathcal{O}}(L_f, \mathcal{G})$ *whose vanishing is a sufficient and necessary condition for the existence of a solution to (08V4).*
(2) *If there exists a solution, then the set of isomorphism classes of solutions is principal homogeneous under* $\mathrm{Ext}^1_{\mathcal{O}}(L_f, \mathcal{G})$.
(3) *Given a solution* X'*, the set of automorphisms of* X' *fitting into (08V4) is canonically isomorphic to* $\mathrm{Ext}^0_{\mathcal{O}}(L_f, \mathcal{G})$.

Proof. Via the identifications $NL_f = \tau_{\geq -1} L_f$ (Lemma 08V2) and $H^0(L_f) = \Omega_f$ (Lemma 08V0) we have seen parts (2) and (3) in Deformation Theory, Lemmas 08UG and 08UK.

Proof of (1). To match notation with Deformation Theory, Section 08UE we will write $NL_f = NL_{\mathcal{O}/\mathcal{O}_\mathcal{B}}$ and $L_f = L_{\mathcal{O}/\mathcal{O}_\mathcal{B}}$ and similarly for the morphisms t and $t \circ f$. By Deformation Theory, Lemma 0GQ9 there exists an element

$$\xi' \in \mathrm{Ext}^1_{\mathcal{O}}(Lf^* \, NL_{\mathcal{O}_\mathcal{B}/\mathcal{O}_{\mathcal{B}'}}, \mathcal{G}) = \mathrm{Ext}^1_{\mathcal{O}}(Lf^* L_{\mathcal{O}_\mathcal{B}/\mathcal{O}_{\mathcal{B}'}}, \mathcal{G})$$

such that a solution exists if and only if this element is in the image of the map

$$\mathrm{Ext}^1_{\mathcal{O}}(NL_{\mathcal{O}/\mathcal{O}_{\mathcal{B}'}}, \mathcal{G}) = \mathrm{Ext}^1_{\mathcal{O}}(L_{\mathcal{O}/\mathcal{O}_{\mathcal{B}'}}, \mathcal{G}) \longrightarrow \mathrm{Ext}^1_{\mathcal{O}}(Lf^* L_{\mathcal{O}_\mathcal{B}/\mathcal{O}_{\mathcal{B}'}}, \mathcal{G})$$

The distinguished triangle of Lemma 08V1 for f and t gives rise to a long exact sequence

$$\ldots \to \mathrm{Ext}^1_{\mathcal{O}}(L_{\mathcal{O}/\mathcal{O}_{\mathcal{B}'}}, \mathcal{G}) \to \mathrm{Ext}^1_{\mathcal{O}}(Lf^* L_{\mathcal{O}_\mathcal{B}/\mathcal{O}_{\mathcal{B}'}}, \mathcal{G}) \to \mathrm{Ext}^1_{\mathcal{O}}(L_{\mathcal{O}/\mathcal{O}_\mathcal{B}}, \mathcal{G})$$

Hence taking ξ the image of ξ' works. □